

Matthew Maloni

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EDUCATION

University of Florida, Herbert Wertheim College of Engineering

M.S. in Electrical and Computer Engineering

Incoming Spring 2027

Gainesville, FL

University of Florida, College of Liberal Arts and Sciences

Bachelor of Science in Data Science

August 2023 – December 2026

Gainesville, FL

GPA: 3.62 / 4.00 · Dean's List: Fall 2023, Fall 2025

Relevant Coursework: Applied Machine Learning Systems; Data Structures & Algorithms; Linear Algebra for Data Science; Probability; Linear Regression; Programming with Data in R; Digital Logic; Information and Database Systems

WORK EXPERIENCE

Kore Capital

May 2026 – Present

Financial Engineering Intern

Boynton Beach, FL

- Co-developed a linear-regression equity factor model spanning momentum, value, quality, size, and growth baskets, applying R^2 and autocorrelation testing on Bloomberg data to evaluate hedge quality and factor attribution.
- Integrated Claude MCP servers into the firm's .NET portfolio system and Excel add-in, condensing tax-lot, PnL, and reporting workflows into automated tools that surface substrategy-level analytics.
- Consolidated 100K+ position, tax-lot, and trade records into a unified analytics layer, building per-campaign visualizations that expose trading patterns and adverse-move risk.

Pastry Art Wine Club

October 2025 – Present

Software Developer

Sarasota, FL (Remote)

- Designed and built a production iOS app for a wine retailer's subscription club, from initial requirements through App Store review and release, fixing production bugs and shipping updates against real customer usage.
- Built a wine-label scanner that reads a user's photo and matches it to the store's catalog in a single AI call, and a recommendation chatbot that builds user memories from how they interact with the app and its AI.

Trustwell

April 2025 – August 2025

Data Science Intern

Salem, OR (Remote)

- Compiled multi-region regulatory datasets and integrated the OpenAI API to improve response quality for Trustwell's food-safety compliance assistant; supported the frontend team's AngularJS-to-Vue-3 migration.

RESEARCH & LEADERSHIP

IEEE Signal Processing Society, UF Student Chapter

September 2025 – Present

Research Board, Project Lead

Gainesville, FL

- Founded and lead a five-member research group building binaural (dual-microphone) audio scene analysis for real-world sound-source localization and separation.
- Implemented scoring and trust-gating algorithms for Plato's Cave, a knowledge-graph paper-verification system; co-authored the resulting paper, accepted to the AAAI Spring Symposium (AAAI-MAKE) 2026.

Nettles Lab, UF Scripps Institute

January 2026 – May 2026

Undergraduate Researcher

Jupiter, FL

- Trained a VAMPnet in PyTorch mapping molecular dynamics trajectories to metastable conformational states for NIH-funded glucocorticoid receptor drug discovery, reaching a VAMP-2 score of 2.87 against a max of 3.0.

Beta Theta Pi, Gamma Xi Chapter

August 2025 – Present

Internal Philanthropy Chair

Gainesville, FL

- Directed Beta Bring It On 2026, raising a record \$64,000 for the Alzheimer's Association and selling out the Phillips Center's 1,700 seats with 300+ participants across 19 teams.

PROJECTS

PortalPath: Graph-Based Ranking for the CFB Transfer Portal

March 2026 · Team of 3 · Python, C++, Dash

- Built an end-to-end pipeline over the CollegeFootballData API condensing 726,000 raw records into 75,492 clean player-seasons; designed a PageRank-based prestige-diffusion model over a 5,201-edge transfer graph ranking 132 programs, deployed as an interactive Dash dashboard.

CNN Medical Image Classification

November 2025 · TensorFlow/Keras, HiPerGator

- Compared four CNN architectures on chest X-ray and diabetic retinopathy classification, achieving 94.77% accuracy and 83.61% AUC across 14 thoracic pathologies over 78,468 images.

SKILLS

Languages: Python, Java, C++, R, SQL, MATLAB, Visual Basic, HTML/CSS, Bash

Machine Learning: PyTorch, TensorFlow/Keras, scikit-learn, deeptime, CNNs, VAMPnets/Markov state models, self-supervised learning, random forests, k-means, regularized regression, cross-validation, PCA/ISOMAP, feature engineering

AI Systems: OpenAI API, Claude/MCP, LLM agent design, retrieval-augmented pipelines, vector databases, knowledge graphs

Infrastructure & Computing: AWS, PostgreSQL, Supabase, Docker, Git/GitHub, HPC (SLURM, HiPerGator)